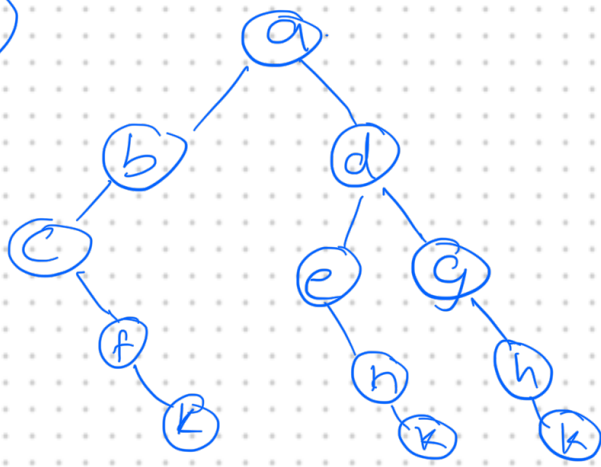


①



For this exercise, we chose the BFS (Breadth-first Search) as we think it's more suitable, because the exercise asks for the minimal depth path.

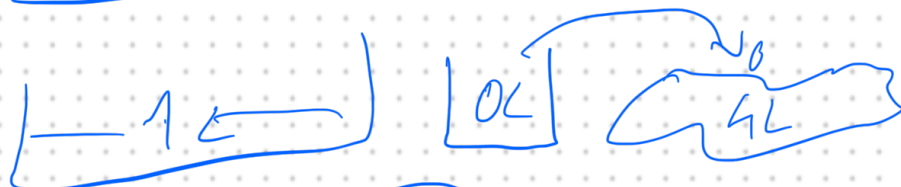
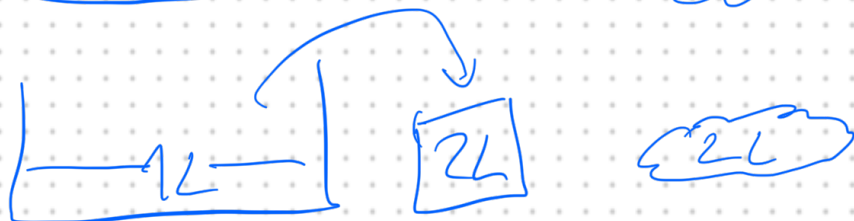
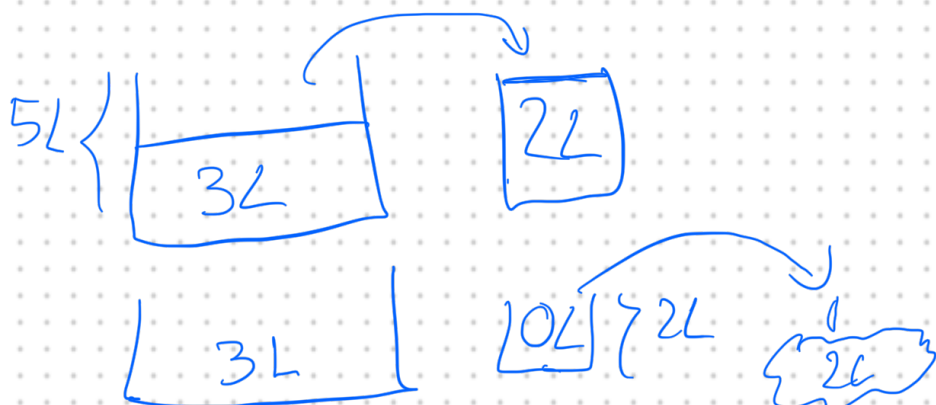
②

t	discovered	Queue
0	a	$\langle a \rangle$
1	b, d	$\langle a, b \rangle ; \langle a, d \rangle$
2	c	$\langle a, d \rangle , \langle a, b, c \rangle$
3	e, g	$\langle a, d, e \rangle ; \langle a, d, g \rangle ; \langle a, b, c \rangle$
4	f	$\langle a, d, e \rangle ; \langle a, d, g \rangle ; \langle a, b, c, f \rangle$
5	h	$\langle a, d, e, h \rangle ; \langle a, d, g \rangle ; \langle a, b, c, f \rangle$
6	k	$\langle a, d, e, h \rangle ; \langle a, b, c, f, k \rangle$

7 return $\langle a, b, c, f, k \rangle$

③

t	Discovered	Queue
0	a	$\langle a \rangle$
1	b, d	$\langle a, b \rangle ; \langle a, d \rangle$
2	c	$\langle a, b, c \rangle ; \langle a, d \rangle$
3	f	$\langle a, b, c, f \rangle ; \langle a, d \rangle$
4	k	$\langle a, b, c, f, k \rangle ; \langle a, d \rangle$
5	return $\langle a, b, c, f, k \rangle$	


$$\begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix}$$


$P = (Z, S_0, S_T)$ $Z = (S, A)$ $x \rightarrow$ amount of water in 5L
 $y \rightarrow$ amount of water in 2L jug
 $S = \{(x, y) \mid 0 \leq x \leq 5 \wedge 0 \leq y \leq 2\}$

$T_{5-2} \rightarrow$ Transfer the 5L to the 2L until it's full or there is no more water in the 5L jug

$P_2 \rightarrow$ Pour the water from the 2L jug

$P_5 \rightarrow$ Pour the water from the 5L jug

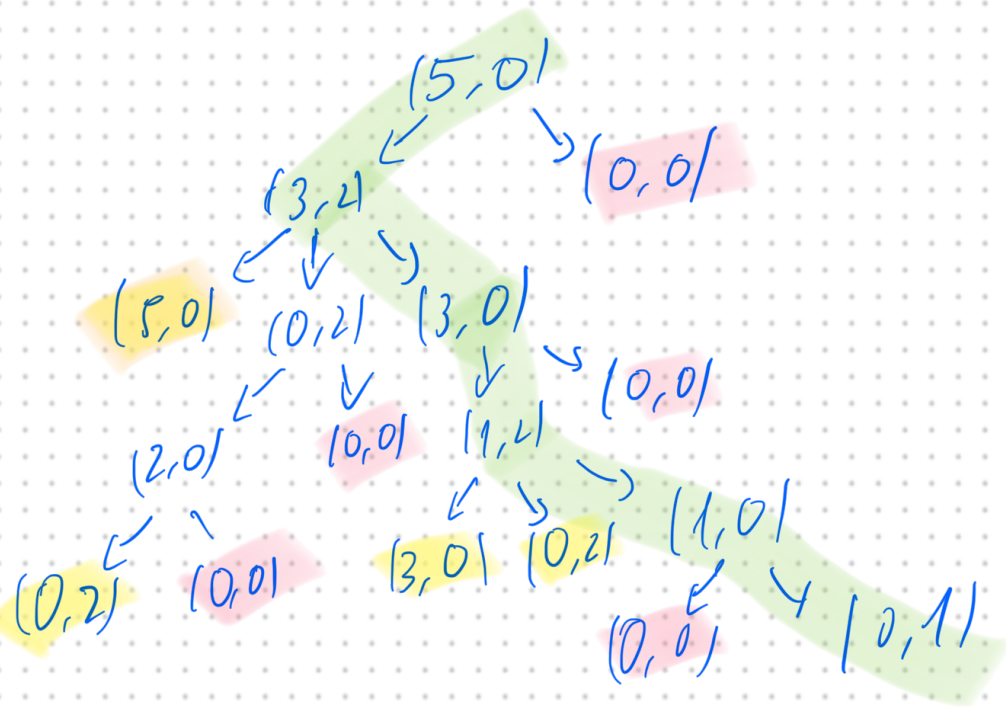
$T_{2-5} \rightarrow$ transfer the 2L to the 5L until there's no more water in the 2L jug

$$A^2 = \{T_{5-2}, T_{2-5}, P_2, P_5\}$$

$$S_0 = (5, 0) \quad S_T = (x, 1)$$

$$(5,0) \xrightarrow{T_{5-2}} (3,2) \xrightarrow{P_2} (3,0) \xrightarrow{T_{5-2}} (1,2) \xrightarrow{P_2} (1,0) \xrightarrow{T_{5-2}} (0,1)$$

Solution path



⑤

$$P = (Z, s_0, s_f) \quad Z = (S, A)$$

$S \rightarrow$ The set of all possible configurations on an $n \times n$ board, where the queens are set so that they cannot threat each other. $0 \leq \text{queens} \leq n$

$A \rightarrow$ place a queen, in the next free column, in a square that doesn't threat other queens

$s_0 \rightarrow$ Empty board

$s_f \rightarrow$ Set of valid configurations with n queens placed.

A state can be represented as a list of row positions $[y_1, y_2, \dots, y_i]$
 $i = n^{\text{th}}$ of the column

As it is a complex problem, the depth-first search is a more efficient approach

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